API

Tuğberk Kocatekin

tugberkkocatekin@arel.edu.tr

API (Application Programming Interface)

- Connects two computers or computer programs through an interface.
  - Connects software pieces and computer with each other
- Simplify programming and can hide a system's internal details
  - Encapsulation
  - Exposes useful parts to programmers
  - You can use file operations, but don't know how it works underneath.
- Requires to build a standard, or document
  - API Specification

Java API (JDBC)

- Standard way for Java programs to communicate with databases.
- Each db vendor has its own protocol for connecting to database.
- JDBC API abstracts all and defines a common ground
  - Now, every Java program can use the same set of functions to talk to any database.
- In short, API in general is provided by the development kit
  - Here, it is JDK
- It consists of a large collection of Java library classes and interfaces.
  - I/O handling, networking, data structures, GUI dev, etc.
  - Enables developers to build applications without reinventing common routines and functionalities.

Web API

- Interfaces that allow different software applications to communicate with each other over the web.
- Defines **endpoints** and **methods** that developers can use to request data or perform actions over the internet.
  - Retrieving user data
  - Sending / Receiving payments, etc.
- Typically built using web technologies and standards
  - REST, SOAP, GraphQL
  - Enable interactions between applications over the web.

Endpoint

- A specific network-accessible address where a client can interact with a service.
- Typically defined by
  - An URI
  - An HTTP Method
- The example on the right bottom is two **different** endpoints.
  - However in daily life we would use **/users** as endpoint.

```
GET /users
POST /users
```

API

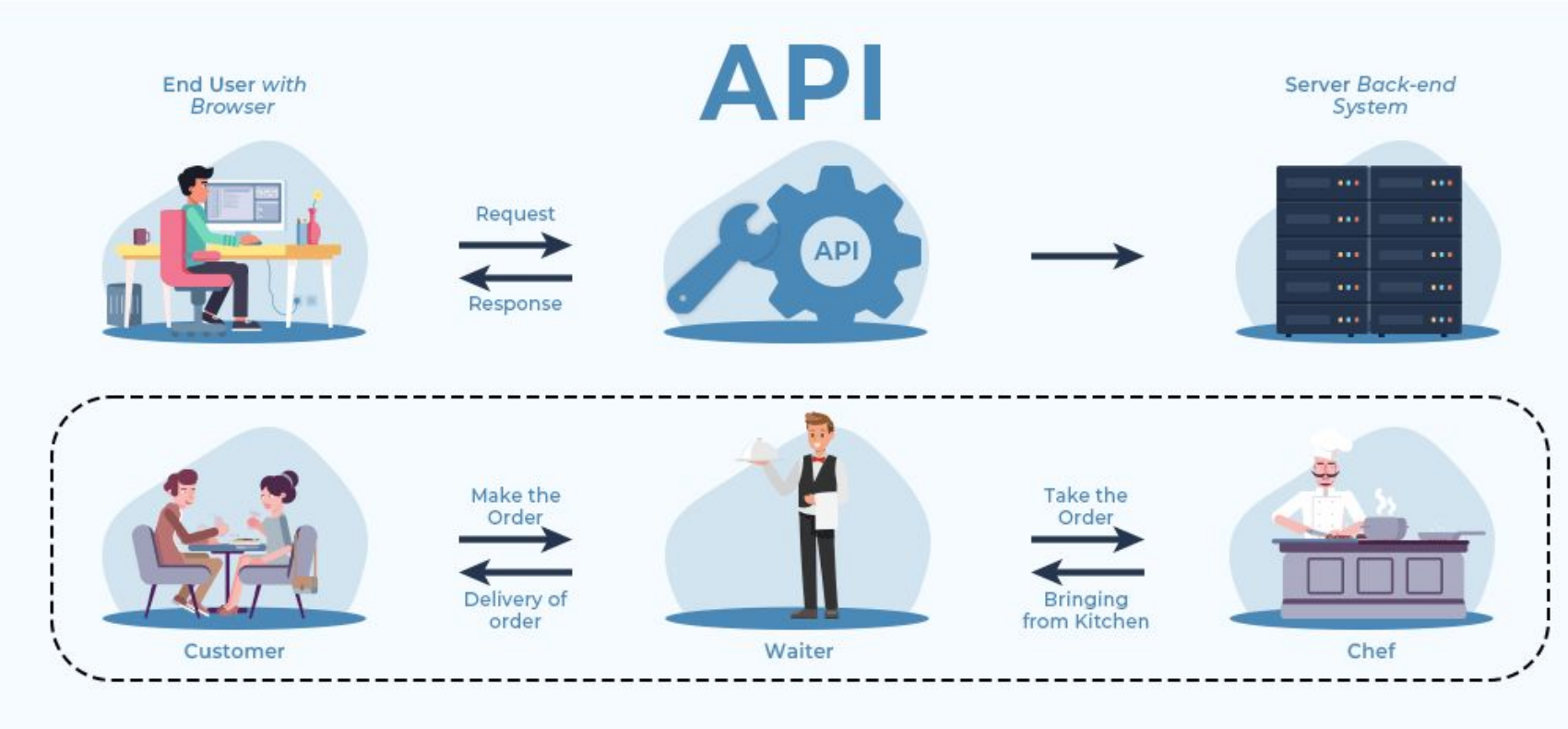
- A contract of service between two applications.
- Defines how the two communicate with each other using
  - **Requests** and **responses**
- Documentation contains information on how developers are to structure those requests and responses.

API

- Client and server
  - Client: Application sending the request
  - Server: Application sending the response
- Several APIs
  - SOAP, RPC, Websocket, REST
  - REST is the most popular
- Client sends requests to the server as data
- Server uses this to start internal functions and returns output data
- Encapsulation
  - We have no idea about the name of the database, schema etc. as the user
  - It encapsulates important data, and backend determines what the user can add / view

REST API

- REST: Representational State Transfer
- Defines a set of functions like
  - GET, PUT, DELETE, etc.
  - Clients can use these to access server data.
  - They exchange data using HTTP.
- Stateless
  - Means server do not save client data between requests.
  - Client requests are similar to URLs we type in our browser.
  - Response from the server is plain data
    - Without any graphical rendering.


API
End User with Browser
Request
Response
API
Server Back-end System
Make the Order
Delivery of order
Take the Order
Bringing from Kitchen
Customer
Waiter
Chef

BACKEND
FRONTEND
API's

Public API

- Public APIs are always there.
- It can be free or paid.
- However, since there are people using it, **the documentation must be perfect.**
  - Good documentation capture users.
- Since there are lots of people using them, you can't do constant changes.
  - You cannot expect people to change their code just because you want them to.
- A good public api should be:
  - Easy to learn
  - Good documentation
  - Hard to use it in a wrong way

Examples

- Weather API
- Human API
- Dog API
- Some of them are going to be more complex because there are more parameters

JSON

- Javascript Object Notation
- A *language independent* data format
  - Used to store and transport data
- Derived from Javascript
- Data serialization
  - Process of converting data structures or objects into a format to be stored and transported.
  - Commonly used in web development for this purpose.
- REST API returns a JSON.

```
{
"employees":[
    {"firstName":"John", "lastName":"Doe"},
    {"firstName":"Anna", "lastName":"Smith"},
    {"firstName":"Peter", "lastName":"Jones"}
]
}
```

I DON'T KNOW WHO JASON IS
AND AT THIS POINT
I'M TOO AFRAID TO ASK
imgflip.com

Playing with APIs

- You can find lots of Public APIs and built programs with them.
  - Some of them will want a key and will give it to you freely, but you'll need the key.
- You can write Python programs, Javascript code in **html** and Javascript code in **nodejs** to play with them.
  - In Python, you need **requests** library.
- In nodejs, you can immediately start sending requests to these apis.
- https://dog.ceo/api/breeds/image/random

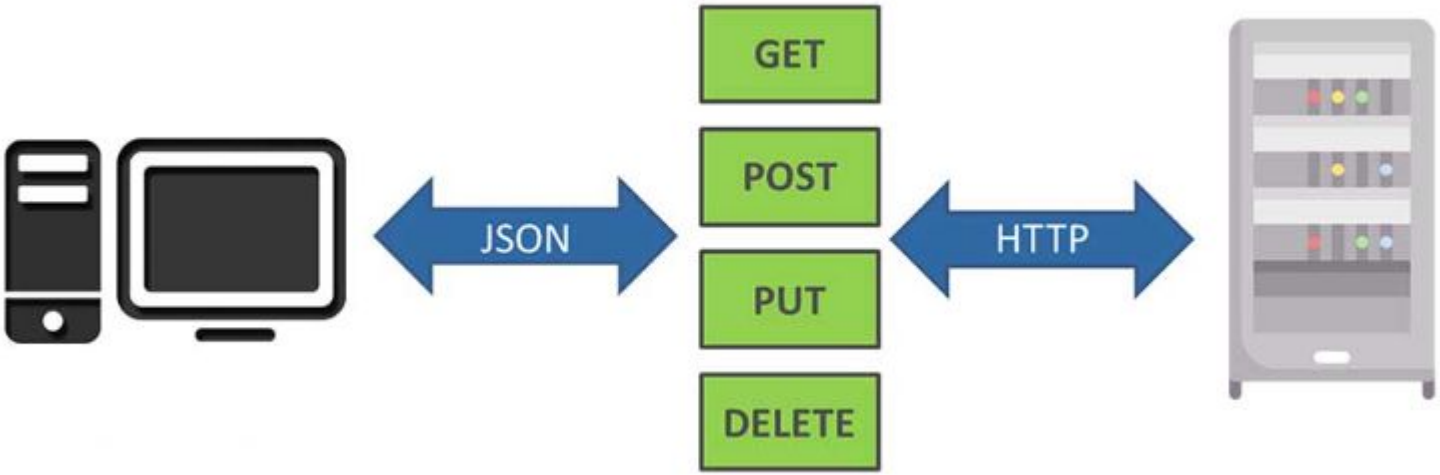
```
1 const apiUrl = "https://dog.ceo/api/breeds/image/random";
2
3 const response = await fetch(apiUrl);
4
5 const data = await response.json();
6
7 console.log(data);
```

Rate Limiting

- Hard stop
  - If the user is over the limit, we stop them with 429.
- Soft stop
  - If user is over the limit, we make them wait. After that time, they can call the API again.
- Throttled stop
  - After a certain limit, you slow down the response time.
- Billable stop
  - After a certain limit, you can charge for money.
  - Can only be used for authenticated users.

API Keys

- Used to authenticate users.
  - If it is fully free, it can overload the system.
- Every time you do an API request, you send your API key.
- If you want to do rate limiting, you need to use authentication.


GET
POST
PUT
DELETE
JSON
HTTP